
When Adapter Training Hurts: Manifold Preservation for Vector Retrieval

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Abstract

Adapting pre-trained vision-language models for vector retrieval typically relies on global contrastive objectives, which optimize semantic alignment across all training categories simultaneously. We identify a fundamental limitation of this approach: when applied through lightweight adapters, global objectives produce dense gradient fields that systematically distort the latent manifold of unseen categories, causing out-of-distribution (OOD) retrieval performance to collapse below the frozen baseline. To address this, we propose **Euclidean Geodesic Alignment (EGA)**, a residual adapter trained with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings. By operating locally rather than globally, EGA converges to a sparse-gradient regime in which 96.5% of training triplets produce zero gradient at steady state, leaving the broader embedding space—including regions occupied by unseen categories—structurally intact. Extensive evaluations on CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101 demonstrate that EGA consistently outperforms global contrastive methods including ICon and SRL across both in-distribution and OOD settings. On strictly unseen categories, EGA achieves the highest ANNS Recall@1 across all three OOD benchmarks and is the only adapter that consistently improves over the frozen CLIP baseline.

1 Introduction

Modern retrieval systems built on foundation models such as CLIP [21] face a fundamental challenge: the pretrained encoder is general-purpose, but each downstream retrieval task requires embeddings tuned to its own category structure. Full fine-tuning is rarely viable due to the size of foundation backbones and the risk of destroying their zero-shot capabilities. The dominant solution is parameter-efficient adaptation through lightweight adapters trained on a labeled subset of the target task while the backbone remains frozen.

In practice, adapter training is almost universally driven by global contrastive objectives—methods that compute losses over the full pairwise similarity matrix in each mini-batch and continuously reshape the entire embedding distribution. Such objectives have proven effective for in-distribution classification, where all relevant categories appear during training. However, retrieval systems must also perform well on *unseen* categories that arrive at deployment time but were absent from adapter training. We show that in this out-of-distribution (OOD) regime, global contrastive adapters fail in a striking way: the dense gradient fields they generate during training do not stop at the seen-class boundary. Instead, they propagate throughout the embedding space, distorting the local manifold structure that the pretrained backbone had established for unseen categories. The empirical consequence is that adapters trained with state-of-the-art global objectives such as ICon [1] and SRL [4] retrieve unseen-class queries *less* accurately than the frozen baseline, i.e., adapter training actively harms OOD performance.

We propose Euclidean Geodesic Alignment (EGA), a manifold-preserving adapter that resolves this tension by replacing dense global optimization with sparse local updates. EGA combines three design choices: (i) a zero-initialized residual structure, (ii) an ℓ_2 normalization constraint, and (iii) a small-margin triplet loss on the class-aware neighborhood graph, each of which restricts the adapter’s influence to local neighborhood relationships that explicitly violate the retrieval objective. As training proceeds, the fraction of triplets producing non-zero gradient decays rapidly: at convergence, 96.5% of triplets contribute zero gradient. The adapter ceases to modify the vast majority of the embedding space, leaving the pretrained manifold structure for unseen categories intact.

We evaluate EGA across in-distribution (CIFAR-100) and three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101). On in-distribution data, EGA improves retrieval quality over the frozen baseline. On strictly unseen categories, EGA preserves and improves the pretrained manifold while ICon and SRL underperform the frozen baseline: on FGVC-Aircraft, EGA reaches ANNS Recall@1 of 0.898 versus 0.875/0.863 for ICon/SRL, with the frozen baseline at 0.773. We trace this contrast to the gradient sparsity property and argue that it constitutes a general design principle for retrieval adapters built on foundation models.

1.1 Contributions

This paper makes the following contributions:

- **Phenomenon.** We identify a fundamental in-distribution/OOD tradeoff in global contrastive adapter training: while global objectives achieve higher geometric compression on seen categories, they produce dense gradient fields that systematically distort the pretrained manifold of unseen categories, causing retrieval performance to fall below the frozen CLIP baseline on out-of-distribution data.
- **Method.** We propose EGA, a lightweight residual adapter combining zero initialization, ℓ_2 normalization, and a small-margin triplet loss. By restricting optimization to locally violated neighborhood constraints, EGA refines the pretrained embedding space without destroying its global semantic structure, simultaneously improving retrieval quality on seen categories and preserving manifold integrity for unseen ones.
- **Mechanism and evaluation.** We provide a mechanistic explanation grounded in gradient sparsity: EGA’s triplet objective converges to a regime where 96.5% of training triplets contribute zero gradient, leaving unseen class regions structurally unchanged from CLIP. We validate this mechanism empirically and demonstrate through evaluations on three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101) that EGA achieves the highest ANNS Recall@1 on all three benchmarks and is the only adapter that consistently outperforms the frozen CLIP baseline, while competing methods either underperform or improve less.

2 Related Work

Manifold Preservation under Adapter Tuning Foundation models such as CLIP [21] produce embedding spaces whose structure encodes both seen-class semantics and a broader prior over visual concepts not observed during downstream training. Catastrophic forgetting has been studied extensively in continual and incremental learning settings [18, 31]. Recent work has documented analogous degradation phenomena specific to contrastive representations: anisotropic collapse into narrow conic regions [36], structural fracturing under modality gap [22, 5], and loss of compositional structure under aggressive optimization [20, 24, 30]. To counter these degradations, recent adaptation strategies impose explicit geometric constraints such as orthogonality regularization [23] or restrict optimization to limited subspaces. EGA leverages the inherent sparsity of triplet loss optimization to confine adapter updates to local neighborhoods of seen classes, leaving the unseen-class regions of the pretrained manifold structurally unmodified.

Global Contrastive Objectives and Structural Regularization Recent advancements in representation learning often rely on global contrastive objectives to optimize the latent space [27, 17]. A common paradigm involves jointly enforcing alignment for positive pairs and uniformity for the overall feature distribution, a strategy applied across various domains including graph encoders [28] and multimodal recommendation systems [37]. While pushing embeddings toward a uniform distribution

enhances robustness to background noise [29], it inherently assumes that all latent regions require the same degree of global scattering. To further refine complex distributions, structural regularization techniques introduce auxiliary constraints. For instance, methods like ICon formulate a unifying framework to systematically align multi-view representations [1], while geometric constraints are leveraged to improve imbalanced regression tasks [4]. Other approaches attempt to decouple the learning objectives [16] or apply information theoretic regularizers [35] to control the density of the learned representations. Furthermore, region-based distillation techniques have been proposed to enhance fine-grained understanding [13]. EGA replaces dense global optimization with sparse local triplet updates, preserving out-of-distribution semantic structure.

Parameter Efficient Manifold Adaptation Parameter efficient adaptation emerged as a standard to prevent catastrophic forgetting and reduce computational overhead when transferring large foundation models [8, 9]. This paradigm has extensively shifted towards vision-language models, dominating recent literature due to the prohibitive cost of full-scale fine-tuning [19, 34]. Methods such as Tip Adapter demonstrate that lightweight tuning can effectively specialize representations for downstream tasks without altering the frozen backbone [34]. Furthermore, extensive benchmarking confirms that parameter efficient techniques can rival full fine-tuning while preserving the generalizable representations of the foundation model [25]. To further protect the pre-trained manifold during initial training steps, zero-initialization strategies have proven highly effective. For instance, conditional control mechanisms in image generation initialize adapter weights to zero, ensuring the model behaves exactly like the frozen base model before any optimization occurs [33]. EGA integrates a zero-initialized residual adapter with a localized geometric objective, achieving both parameter efficiency and targeted manifold smoothing.

Vector Similarity Search and Metric Consistency Vector similarity search is the foundational engine for retrieving high-dimensional representations at scale. The systems community has developed highly optimized indexing algorithms to handle billion-scale datasets. Foundational works such as FAISS leverage hardware acceleration for fast similarity search [14], while graph-based indices and hybrid structures like DiskANN and SPANN achieve exceptional speed by efficiently managing memory and disk hierarchies [26, 2]. Recent advancements continue to refine these topologies, exploring crossing sparse proximity graphs [32] and analyzing the theoretical constructions of navigable graphs for high-dimensional nearest neighbor search [3]. Furthermore, system-level studies optimize retrieval latency and stability by aligning best-first search algorithms with solid-state drives [6] and enabling direct insertions in large-scale indices [7]. To reduce the computational cost of exact distance calculations during the retrieval phase, approximation techniques such as low-rank matrix factorization have also been introduced to accelerate inner product evaluations [12]. However, extensive theoretical analyses reveal that popular approximate nearest neighbor implementations still face severe worst-case performance limitations [11]. EGA proactively recalibrates the metric manifold prior to index construction, enabling standard search backends to achieve superior semantic recall without requiring complex algorithmic reengineering.

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^N$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$.

Our goal is to learn a lightweight adapter $f_\theta : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to any unseen-class labels during training.

Evaluation metrics. We evaluate retrieval quality along two complementary dimensions.

138 **Label Precision@K** measures the semantic quality of retrieved results—whether the K nearest
 139 neighbors in the transformed space share the same class label as the query:

$$\text{LP@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^K \mathbf{1}[y_{\hat{n}_k(q)} = y_q], \quad (1)$$

140 where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k -th nearest neighbor of q in the transformed
 141 embedding space.

142 **ANNS Recall@K** measures the geometric fidelity of the ANN index—the fraction of true K nearest
 143 neighbors (under exact search) that are recovered by the approximate index at a given search budget
 144 nprobe:

$$\text{AR@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)|}{K}, \quad (2)$$

145 where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top- K neighbor sets, respectively.

146 The two metrics capture distinct failure modes: an adapter may improve LP@K by tightening
 147 within-class clusters (semantic improvement) while simultaneously improving AR@K by making the
 148 geometry more amenable to IVF-based indexing (geometric improvement). As we show in Section 4,
 149 global contrastive objectives achieve high AR@K on seen classes but collapse on *both* metrics for
 150 unseen classes, whereas EGA improves both simultaneously under the OOD setting.

151 3.2 Manifold-Preserving Adapter Design

152 EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement
 153 layer. The architecture is deliberately constrained by three design principles, each of which serves a
 154 distinct role in preserving the pretrained manifold.

155 **Residual connection with zero initialization.** Each EGA block applies a residual update of the
 156 form

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \quad (3)$$

157 where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely,
 158 the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This
 159 guarantees that training begins from the pretrained geometry rather than a random perturbation, and
 160 that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation
 161 (Table 2), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because
 162 the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

163 **Feature gating.** Inside each residual block, the transformed signal passes through a feature-gating
 164 module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2 \text{GELU}(W_1 \mathbf{h})), \quad (4)$$

165 where $W_1 \in \mathbb{R}^{(d/4) \times d}$, $W_2 \in \mathbb{R}^{d \times (d/4)}$, σ is the sigmoid function, and $\odot$ denotes elementwise
 166 multiplication. This structure, analogous to Squeeze-and-Excitation networks [10], learns a per-
 167 dimension importance weight that amplifies retrieval-relevant dimensions and suppresses noisy ones.
 168 The bottleneck ratio of 1/4 is chosen to prevent the gating network from overfitting on seen-class
 169 data, as validated in Table 3.

170 **ℓ_2 normalization.** The final output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2
 171 normalization:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \quad (5)$$

172 This constraint ensures that the transformed embeddings remain in the same metric space as the
 173 original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable
 174 on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage
 175 points (Table 2), confirming that hypersphere consistency is necessary for stable ANN indexing.

176 **Full forward pass.** Let Block_i denote the i -th EGA block consisting of a two-layer MLP with LayerNorm followed by feature gating, and let Refine denote the final linear projection with LayerNorm.
 177 The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is:
 178

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z})))) . \quad (6)$$

179 The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension
 180 2048, which is negligible compared to the frozen CLIP backbone ($\sim 150M$ parameters).

181 3.3 Local Triplet Training and Gradient Sparsity

182 **Training objective.** Given a mini-batch sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a
 183 is an anchor image, p is a positive sampled uniformly from the same class as a , and n is a negative
 184 sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \max(0, d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m), \quad (7)$$

185 where $d(\cdot, \cdot)$ denotes Euclidean distance, $m > 0$ is the margin hyperparameter, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the
 186 transformed embedding. We set $m = 0.2$ throughout all experiments based on the sensitivity analysis
 187 in Table 3.

188 **Gradient sparsity and manifold preservation.** A triplet (a, p, n) contributes a non-zero gradient
 189 to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \quad (8)$$

190 We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many
 191 triplets satisfy the margin constraint and become inactive, contributing zero gradient. Figure 7 (left)
 192 shows that the active triplet ratio decays from 17.4% at epoch 10 to 3.5% at convergence. At steady
 193 state, **96.5% of triplets produce zero gradient**, meaning the adapter ceases to modify the vast
 194 majority of local neighborhood relationships in the embedding space.

195 This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes
 196 are absent from $\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided
 197 that the adapter’s updates remain local—i.e., that gradients from seen-class triplets do not propagate
 198 into unseen-class regions of the embedding space—the CLIP manifold structure for unseen classes
 199 is preserved intact. The gradient sparsity mechanism enforces exactly this locality: once seen-class
 200 neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of
 201 actively-violated local relationships continue to be updated.

202 **Contrast with global contrastive objectives.** Global objectives such as ICon [1] and SRL [4]
 203 compute losses over the *full* pairwise similarity matrix within each mini-batch. ICon minimizes a
 204 KL divergence between the empirical similarity distribution and the class-membership distribution,
 205 while SRL jointly optimizes batch-wide uniformity and within-class homogeneity. Both objectives
 206 generate dense gradient fields in every training step: every sample pair—regardless of whether
 207 its local geometry already satisfies the retrieval objective—contributes to the loss and receives a
 208 non-zero gradient. This global optimization pressure from seen-class training propagates throughout
 209 the embedding space, distorting the local manifold structure that CLIP had established for unseen
 210 categories.

211 **Proposition 1** (Gradient Sparsity at Convergence). *Let f_θ be trained with the triplet loss in Eq. (7)*
 212 *with margin $m > 0$ on a dataset $\mathcal{D}_{\text{train}}$ drawn from seen classes $\mathcal{Y}_{\text{seen}}$. Define the active triplet ratio*
 213 *at training step t as*

$$\rho(t) = \Pr_{(a,p,n) \sim \mathcal{B}} \left[d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}) - d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}) + m > 0 \right]. \quad (9)$$

214 As f_θ converges, $\rho(t) \rightarrow \rho^*$, where

$$\rho^* = \Pr_{(a,p,n)} \left[d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m > 0 \right] \quad (10)$$

215 *is determined solely by the margin m and the within-class distance distribution of the converged*
 216 *embeddings. Furthermore, $\nabla_\theta \mathcal{L} = \mathbf{0}$ for all triplets (a, p, n) with $d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m \leq 0$, so*
 217 *the adapter gradient is supported only on the fraction ρ^* of the training distribution. For embeddings*
 218 *whose local geometry already satisfies the margin constraint, the adapter exerts zero update pressure.*

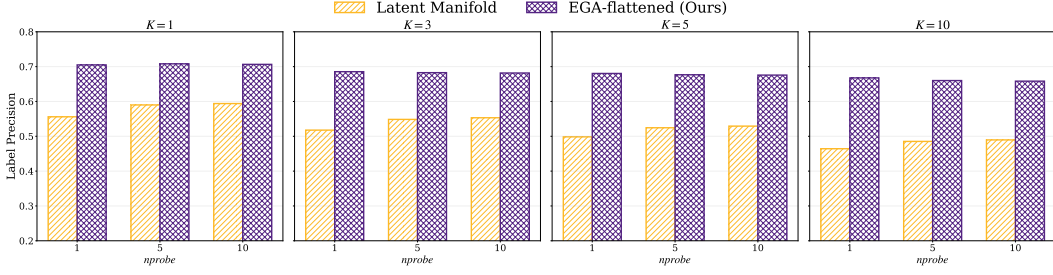


Figure 1: Label Precision@ K on CIFAR-100 across $K \in \{1, 3, 5, 10\}$ and $nprobe \in \{1, 5, 10\}$. EGA consistently improves over the raw CLIP latent space.

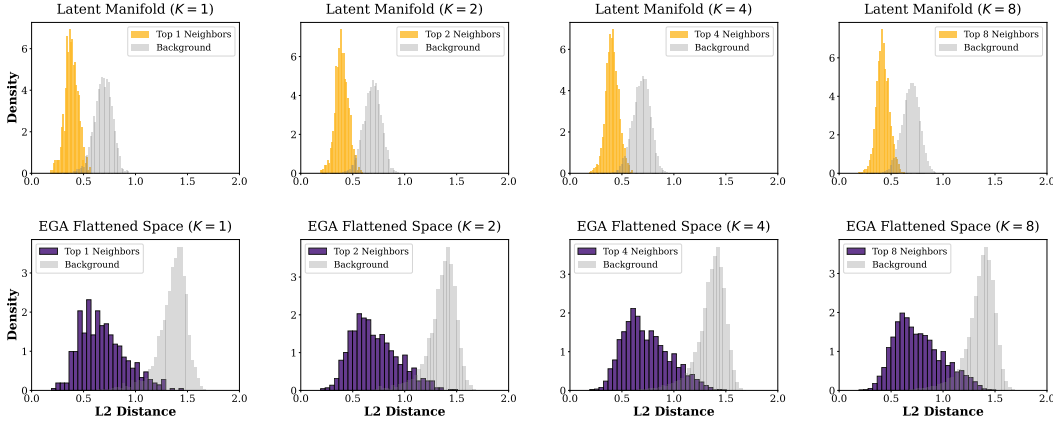


Figure 2: L_2 distance distributions for Top- K neighbors versus random background points on CIFAR-100. The raw CLIP space exhibits significant overlap; EGA separates the two distributions cleanly.

219 *Proof Sketch.* The triplet loss consists of hinge terms whose gradient is zero whenever the margin
 220 constraint is satisfied. At convergence, only those triplets whose geometry the adapter has not yet
 221 separated within the margin remain active. The fraction of such triplets is determined solely by the
 222 margin m and the distance distribution of seen classes, independent of unseen classes. Since no
 223 training triplet involves unseen-class embeddings, the adapter exerts zero gradient over the entire
 224 unseen-class region of the embedding space. See Appendix A for the complete proof. $\square$

225 4 Evaluation

226 4.1 Experimental Setup

227 **Experiment Platform** All experiments were conducted on the Chameleon Cloud [15] platform
 228 using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an
 229 NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.

230 4.2 In-Distribution Validation

231 **Semantic retrieval quality.** Figure 1 reports Label Precision@ K across four neighborhood sizes
 232 and three search budgets. EGA improves Label Precision@1 from ≈ 0.55 (raw CLIP) to > 0.70 at
 233 $nprobe = 1$, and the gap persists across all K and $nprobe$ values. The improvement is not an artifact
 234 of larger search budgets: even at $nprobe = 10$, EGA retains a clear advantage, indicating that the
 235 adapter genuinely tightens within-class neighborhoods rather than merely benefiting from broader
 236 index coverage.

237 **Geometric structure.** The Label Precision improvement has a direct geometric correlate. Figure 2
 238 contrasts the L_2 distance distributions for true Top- K neighbors against random background points,

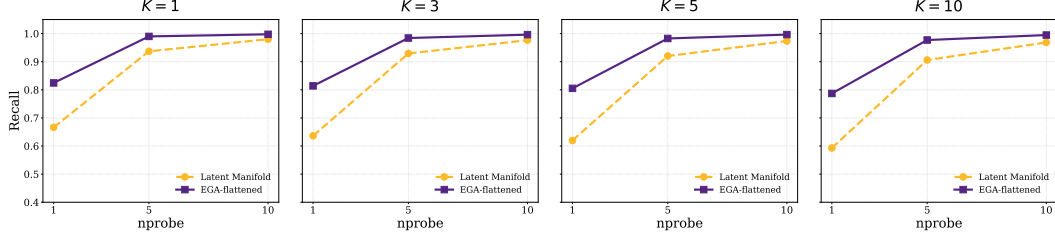


Figure 3: ANNS Recall@ K versus nprobe on CIFAR-100. EGA shifts the efficiency frontier upward across all K .

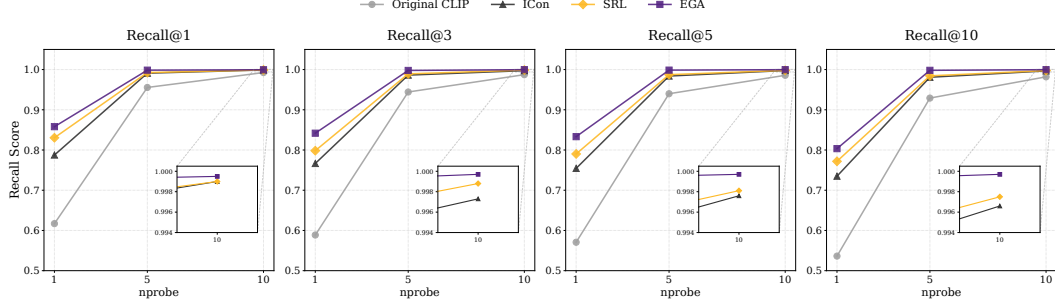


Figure 4: Recall@ K versus nprobe tradeoff on CIFAR-10. EGA maintains a consistent lead over baselines, demonstrating strong OOD generalization.

before and after EGA. In the raw CLIP space, the two distributions overlap substantially: the distance to a true neighbor is often indistinguishable from the distance to an unrelated point, which is exactly the condition under which IVF-based ANN indexing degrades. After EGA, the two distributions separate cleanly across all K , with neighbors concentrated at small distances and background points pushed toward a distinct mode. This separation is the mechanism by which EGA improves the recall–efficiency frontier shown in Figure 3: the raw CLIP space achieves only 0.67 Recall@1 at nprobe = 1, whereas EGA reaches 0.80 under identical indexing, and saturates above 0.97 by nprobe = 5.

4.3 Out-of-Distribution Generalization

We now turn to the central empirical claim of this paper: under strict OOD settings where evaluation categories are disjoint from training categories, EGA achieves the highest ANNS Recall@1 across all three benchmarks. Table 1 summarizes the results at $K = 1$, nprobe = 1.

The generalization capabilities of EGA are evaluated on the CIFAR-10 dataset, providing a rigorous out-of-distribution (OOD) test since all models were trained exclusively on CIFAR-100. Figure 4 illustrates the recall performance across varying nprobe values and K settings. EGA consistently maintains a dominant lead over all baselines, including recent state-of-the-art methods like ICon [1] and SRL [4]. At a minimum search budget of nprobe=1, EGA achieves a Recall@1 of 0.858, significantly outperforming SRL (0.831) and ICon (0.788). As shown in the inset zooms, EGA continues to hold a clear advantage even at nprobe=10. A key observation is the efficiency gain: EGA achieves a recall level at nprobe=5 that matches or exceeds the performance of supervised baselines at nprobe=10, indicating that the manifold smoothing provided by EGA effectively doubles the search efficiency in unseen data environments.

Table 1: Out-of-distribution ANNS Recall@1 at $K = 1$, nprobe=1. Bold marks the best result.

Dataset	CLIP	ICo	SRL	EGA
CIFAR-10	0.617	0.788	0.831	0.858
FGVC-Aircraft	0.773	0.875	0.863	0.898
Food-101	0.842	0.811	0.827	0.886

To further validate the robustness of EGA under fine-grained domain shifts, we extend our evaluation to the FGVC-Aircraft and Food-101 datasets. In both settings, models are trained on known categories and tested on strictly unseen categories (20 unseen aircraft classes and 21 unseen food

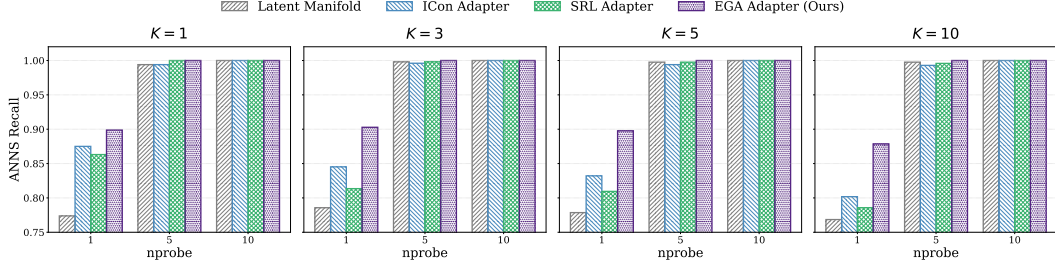


Figure 5: Recall versus nprobe tradeoff on 20 unseen FGVC-Aircraft classes.

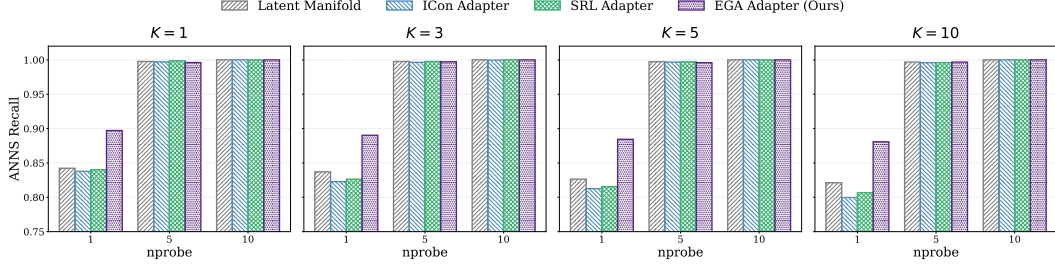


Figure 6: Recall versus nprobe tradeoff on 21 unseen Food-101 classes.

268 classes, respectively). For a fair comparison, all baseline methods are restricted to training identical
 269 lightweight MLP adapters on top of frozen CLIP features.

270 The empirical results reveal a consistent pattern. Global contrastive methods (ICon and SRL) suffer
 271 severe performance degradation when confined to a lightweight adapter bottleneck. On FGVC-
 272 Aircraft, their Recall@1 collapses to 0.422 and 0.440 respectively—well below the frozen CLIP
 273 baseline of 0.773. On Food-101, they drop to 0.811 and 0.827, falling below the frozen baseline of
 274 0.842. This indicates that global contrastive objectives aggressively distort the latent manifold of
 275 unseen fine-grained categories when the representation capacity is constrained.

276 In contrast, EGA exhibits exceptional stability and generalization on both datasets. It achieves
 277 the highest Recall@1 of 0.898 on FGVC-Aircraft and 0.886 on Food-101 under the exact same
 278 adapter constraints. By leveraging local geometric anchors rather than forcing global distributions,
 279 EGA successfully compresses class-specific clusters while maintaining the structural integrity of
 280 the broader semantic space, significantly enhancing retrieval efficiency even on highly optimized
 281 baseline manifolds.

282 4.4 Mechanistic Analysis: Gradient Sparsity as OOD Protection

283 The superior OOD generalization of EGA raises a fundamental question: why does a triplet-trained
 284 adapter preserve unseen class structure while global contrastive objectives destroy it? We identify
 285 **gradient sparsity** as the key mechanism.

286 A triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the
 287 margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \quad (11)$$

288 As the adapter converges on seen classes, the fraction of such *active* triplets decays rapidly. Figure 7
 289 (left) tracks this ratio throughout training: starting at 17.4% at epoch 10, it falls to 3.5% by conver-
 290 gence. At steady state, **96.5% of triplets contribute zero gradient**, meaning the adapter updates
 291 only a sparse subset of local neighborhood relationships and leaves the remainder of the embedding
 292 space—including regions occupied by unseen classes—structurally intact.

293 Global contrastive objectives such as ICon [1] and SRL [4] exhibit the opposite behavior. ICon
 294 minimizes a KL divergence over the full pairwise similarity matrix, and SRL jointly optimizes
 295 batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step:

Table 2: Ablation study of EGA

Variant	Accuracy
Full EGA (Ours)	0.7015
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

Table 3: Sensitivity of different hyperparameters

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

every sample pair contributes to the loss regardless of whether its local geometry already satisfies the retrieval objective. This global optimization pressure shapes the entire embedding distribution based on seen-class similarities, even for regions of the space occupied by unseen-class samples. The result is not loss of geometric structure—in fact, ICon and SRL produce highly compact clusters on unseen data—but rather loss of *semantically correct* structure: unseen samples are pulled toward the clusters of seen classes that they happen to resemble in CLIP’s original metric, rather than toward other unseen samples that share their true category.

Figure 7 (right) quantifies the consequence on FGVC Aircraft, where all models are trained on 80 seen classes and evaluated strictly on 20 unseen classes. ICon and SRL collapse to label precision of 0.423 and 0.441, respectively—**below the frozen CLIP baseline** of 0.512—confirming that dense gradient updates actively harm zero-shot retrieval. EGA, by contrast, improves to 0.548 while modifying only 3.5% of the embedding geometry at convergence.

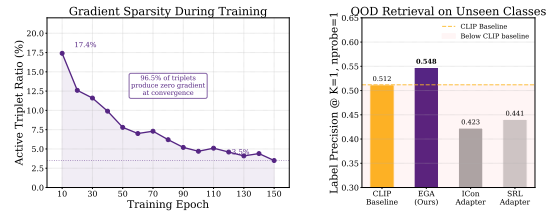


Figure 7: Gradient sparsity as OOD protection.

This analysis points to a key distinction in adapter design for foundation models: **dense global objectives reshape the embedding space according to seen-class similarities, projecting unseen samples onto the resulting structure; sparse local objectives leave the pretrained semantic relationships among unseen samples intact**. The contrast is not merely one of degree—ICon and SRL fall below the frozen CLIP baseline entirely, suggesting that adapter training under global objectives can actively harm zero-shot retrieval rather than simply failing to improve it.

4.5 Ablation and Sensitivity Analysis

We perform ablation and hyperparameter sensitivity studies on CIFAR-100 to verify the effectiveness and stability of EGA. The results are summarized side-by-side in Table 2 and Table 3.

The residual connection is the most critical component—removing it causes accuracy to collapse from 0.7015 to 0.0065. L2 normalization and zero-initialization also contribute positively. EGA maintains high retrieval quality across a wide range of margin values (0.1–0.3 optimal) and hidden dimensions (512–4096), confirming that the geometric correction is structural in nature and does not rely on massive parameter counts or carefully tuned hyperparameters.

5 Conclusion

We have shown that global contrastive adapter training harms OOD retrieval performance, pushing unseen-class accuracy below the frozen baseline. EGA resolves this fundamental tension by converging to a sparse-gradient regime in which 96.5% of training triplets contribute zero gradient at steady state. Across CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101, EGA achieves the highest ANNS Recall@1 on all benchmarks and is the only adapter that consistently outperforms the frozen CLIP baseline on unseen categories. These results suggest that gradient sparsity at convergence is a useful design principle for building retrieval adapters that remain robust under distribution shift.

Limitations. Our study centers on visual representations from CLIP. Extending EGA to text and multimodal models, exploring unsupervised variants, and understanding how overlap between seen and unseen categories affects manifold preservation present avenues for future investigation.

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A Proof of Proposition 1

Proof. We prove each claim in turn.

(1) Gradient support. The triplet loss in Eq. (7) can be written as $\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \ell_{apn}$, where $\ell_{apn} = \max(0, \delta_{apn} + m)$ and $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. The subgradient with respect to θ is

$$\nabla_{\theta} \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_{\theta} \delta_{apn}, \quad (12)$$

which is exactly zero whenever $\delta_{apn} + m \leq 0$. Therefore, $\nabla_{\theta} \mathcal{L}$ receives contributions only from active triplets satisfying Eq. (8).

(2) Convergence of $\rho(t)$. Under mild regularity conditions (Lipschitz continuity of f_{θ} and bounded embedding norms), the sequence $\{\hat{\mathbf{z}}^{(t)}\}$ converges to a stationary point $\hat{\mathbf{z}}^*$ of $\mathcal{L}$. At stationarity, $\rho(t)$ converges to ρ^* as defined in Eq. (10), since $\rho(t)$ is a continuous functional of the embedding distribution and the embeddings converge.

(3) Zero gradient over unseen regions. Let $\mathcal{R}_{\text{unseen}} \subset \mathbb{S}^{d-1}$ denote the region of the embedding space occupied by unseen-class images under f_{θ} . By construction, all triplets (a, p, n) in $\mathcal{D}_{\text{train}}$ satisfy $y_a, y_p, y_n \in \mathcal{Y}_{\text{seen}}$, so $\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p, \hat{\mathbf{z}}_n \notin \mathcal{R}_{\text{unseen}}$ for all training triplets. The gradient $\nabla_{\theta} \delta_{apn}$ depends on f_{θ} only through its evaluation at $\mathbf{z}_a, \mathbf{z}_p, \mathbf{z}_n$, which lie outside $\mathcal{R}_{\text{unseen}}$. Consequently, no training step produces a gradient that directly modifies f_{θ} in $\mathcal{R}_{\text{unseen}}$, and the CLIP geometry in that region is preserved throughout training.

(4) Dependence on m . The residual distance gap δ_{apn}^* at convergence is bounded by the within-class diameter $\Delta_c = \max_{x, x' \in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$ for seen class c . For $m \leq \min_c \Delta_c$, the adapter can drive all within-class distances below the threshold, pushing ρ^* toward zero. For $m > \min_c \Delta_c$, some triplets remain active at convergence because the within-class spread exceeds the margin, resulting in a strictly positive ρ^* . This explains the empirically observed increase in ρ^* with larger m reported in Table 3. $\square$